4.7.3 Acids and alkalis

Some chemical substances can be classified as acids or alkalis. Characterising substances in this way helps to make sense of how chemicals react together, to establish patterns and to make predictions about chemical changes. This topic provides opportunities to write chemical formulae and equations, and apply the quantitative methods from Chemical quantities (page 89). Practical content includes methods used to prepare and purify soluble salts. Theoretical content includes how to describe the energy changes associated with neutralisation reactions and how ionic theory can account for the similarities in the reactions of acids.

There are two required practicals: one is the preparation of a pure, dry sample of a soluble salt from an insoluble substance. The other investigates the variables that affect temperature changes in solutions.

4.7.3.1 Reactions of acids

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall that acids react with some metals and with carbonates, and write equations predicting products from given reactants. Describe tests to identify selected gases including hydrogen and carbon dioxide.	Acids react with some metals to produce salts and hydrogen. Knowledge of reactions with metals is limited to those of magnesium, zinc and iron with hydrochloric and sulfuric acids. The test for hydrogen uses a burning splint held at the open end of a test tube of the gas. Acids react with metal carbonates to produce salts, water and carbon dioxide. The test for carbon dioxide uses an aqueous solution of calcium hydroxide (limewater).	WS 1.2 (HT only) Explain, in terms of gain or loss of electrons, that the reactions of metals with acids are redox reactions (see Atoms into ions and ions into atoms (page 134)). WS 4.1 (HT only) Identify which species are oxidised and which are reduced in given chemical reactions.

4.7.3.2 Making salts

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe neutralisation as acid reacting with alkali to form a salt plus water. Describe, explain and exemplify the processes of filtration and crystallisation. Suggest suitable purification techniques given information about the substances involved.	Acids are neutralised by alkalis (eg soluble metal hydroxides) and bases (eg insoluble metal hydroxides and metal oxides) to produce salts and water, and by metal carbonates to produce salts, water and carbon dioxide. The particular salt produced in any reaction between an acid and a base or alkali depends on: • the acid used (hydrochloric acid produces chlorides, nitric acid produces nitrates and sulfuric acid produces sulfates) • the positive ions in the base, alkali or carbonate. Soluble salts can be made from acids by reacting them with solid insoluble substances such as metals, metal oxides, hydroxides or carbonates. The solid is added to the acid until no more reacts, and the excess solid is filtered off to produce a solution of the salt. Salt solutions can be crystallised to produce solid salts.	WS 1.2 Predict products from given reactants. Use the formulae of common ions to deduce the formulae of salts.

Required practical activity 17: preparation of a pure, dry sample of a soluble salt from an insoluble oxide or carbonate, using a Bunsen burner and a water bath or electric heater to evaporate the solution.

AT skills covered by this practical activity: chemistry AT 2, 3, 4 and 6.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in Key opportunities for skills development (page 174).

4.7.3.3 Energy changes and reactions

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Distinguish between endothermic and exothermic reactions on the basis of the temperature change of the surroundings.	When chemical reactions occur, energy is transferred to or from the surroundings. Energy is conserved in chemical reactions. The total amount of energy in the reaction mixture and its surroundings at the end of a chemical reaction is the same as it was at the start. An exothermic reaction is one that gives out energy. This heats up the reaction mixture. Energy then transfers to the surroundings as the reaction mixture then cools. Neutralisation of an acid with an alkali is an example of an exothermic reaction. An endothermic reaction is one that that takes in energy. This cools the reaction mixture. Energy then transfers from the surroundings as the reaction mixture then warms up again. The reaction of citric acid and sodium hydrogen carbonate is an example of an endothermic reaction.	WS 1.2 Identify examples of exothermic and endothermic reactions based on the temperature change of the reaction mixture.

Required practical activity 18: investigate the variables that affect the temperature changes of a series of reactions in solutions, eg acid plus metals, acid plus carbonates, neutralisations, displacement of metals.

AT skills covered by this practical activity: chemistry AT 1, 3, 5 and 6.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in Key opportunities for skills development (page 175).

4.7.3.4 The pH scale and neutralisation

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall that acids form hydrogen ions when they dissolve in water and solutions of alkalis contain hydroxide ions. (HT only) Use the formulae of common ions to write balanced ionic equations. Recall that relative acidity and alkalinity are measured by pH. Recognise that aqueous neutralisation reactions can be generalised to hydrogen ions reacting with hydroxide ions to form water.	Acids produce hydrogen ions (H^+) in aqueous solutions. Aqueous solutions of alkalis contain hydroxide ions (OH^-). The pH scale, from 0 to 14, is a measure of the acidity or alkalinity of a solution and can be measured using universal indicator or a pH probe. A solution with pH 7 is neutral. Aqueous solutions of acids have pH values of less than 7 and aqueous solutions of alkalis have pH values greater than 7. In neutralisation reactions between an acid and an alkali, hydrogen ions react with hydroxide ions to produce water.	WS 1.2 Write an ionic equation to represent neutralisation. WS 2.3 Describe the use of universal indicator or a wide range indicator to measure the approximate pH of a solution. WS 3.2 Use the pH scale to identify acidic or alkaline solutions.

4.7.3.5 Strong and weak acids (HT only)

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Use and explain the terms dilute and concentrated (amount of substance) and weak and strong (degree of ionisation) in relation to acids. Recall that as hydrogen ion concentration increases by a factor of ten the pH value of a solution decreases by a factor of one. Describe neutrality and relative acidity and alkalinity in terms of the effect of the concentration of hydrogen ions on the numerical value of pH (whole numbers only).	A strong acid is completely ionised in aqueous solution. Examples of strong acids are hydrochloric, nitric and sulfuric acids. A weak acid is only partially ionised in aqueous solution. Examples of weak acids are ethanoic, citric and carbonic acids. For a given concentration of aqueous solutions, the stronger an acid, the lower the pH.	